

Lab 0 – The Chromatin Game

A first, hands-on look at Hi-C maps and chromatin folding

Warm-up for Laboratory 1 · Time needed: 20–25 minutes

Hi-C experiments give an $N \times N$ **contact map**: entry (i, j) counts how often loci i and j touch in 3D. This lab has you fold a virtual chromosome yourself and watch that map emerge, so the idea stops being abstract.

Task 1. Set up an isolated Python environment and install the game. The game is open-source: <https://github.com/SFGLab/TheChromatinGame>.

```
git clone https://github.com/SFGLab/TheChromatinGame.git
cd TheChromatinGame
python -m venv chromatin_env          # create the environment
source chromatin_env/bin/activate    # on Windows: chromatin_env\Scripts\activate
pip install -r requirements.txt
python main.py
```

A dedicated environment per project – here via venv, or pyenv/conda if you prefer – keeps packages from clashing, a habit worth starting on day one.

Task 2. Open Chromatin MiNI-Lab from the main menu and freely paint compartments, tie a loop, and move a slider. Press C and drag across the colour ribbon to mark beads red (A) or blue (B); press L and click two beads (or a cell on the map) to tie a loop between them; then drag any slider – e.g. temperature or attraction strength – under the heatmap. Everything else is explained in-game (press H). Watch the 3D structure and the heatmap respond together, live: what you are doing here is the easy direction, the one physics is good at, where your choices fix the structure and the structure fixes the map.

Task 3. Go to Single Player, level 1 (normal mode), and reproduce the target heatmap shown to you. A target map is displayed next to your own, live one; using the same painting and loop-tying controls as before, try to make the two look alike, then press Enter to freeze the simulation and score your match. Now you are running the direction biology actually hands you: given the map, recover the structure that produced it. Many different structures can cast nearly the same shadow, and noise can point you toward the wrong one – which is why this is a genuine **inverse problem**, not just a puzzle.

Task 4. Replay the same level in hard mode, where the target's loop anchors and compartment colours are hidden and only its raw contact map is shown. Study that map for a moment – where is the checkerboard strongest, where do bright squares sit on the diagonal – before placing any loops or colour, then score yourself as before. This is the realistic case: a real Hi-C analyst never sees the ground truth, only the measurement. Normal mode had handed you the answer key; hard mode does not, exactly like a real dataset on your desk.

Task 5. Before you leave, write one or two sentences answering: what produced the checkerboard pattern, what produced the bright square on the diagonal, and why was hard mode harder? The same logic – indirect measurement, a physical model, an inferred structure – underlies protein structure determination by X-ray, NMR, or cryo-EM, and it is exactly what we build explicitly, for proteins and chromatin alike, in the *Molecular Simulations* block later in this course. This game is your first playable encounter with that idea.